

A Machine Learning Model for the Early Prediction of Cardiovascular Disease in Patients

Abhishek

Department of Computer Science and
Engineering

Sant Longowal Institute of Engineering
and Technology, Sangrur

abhiunite4plus@gmail.com

Hutashan Vishal Bhagat

Department of AI & ML (CSE)
B V Raju Institute of Technology,

Hyderabad, India

hutashan20@gmail.com

Manminder Singh

Department of Computer Science and
Engineering

Sant Longowal Institute of Engineering
and Technology, Sangrur

manminderldh@gmail.com

Abstract—Cardiovascular disease is a major health concern worldwide, with an estimated 17.9 million deaths each year. It encompasses a range of conditions affecting the heart and blood vessels, including coronary heart disease, stroke, and heart failure. These diseases are often associated with a range of modifiable and non-modifiable risk factors, including high blood pressure, high cholesterol, smoking, physical inactivity, obesity, diabetes, and genetic predisposition. Data analysis and machine learning have emerged as powerful tools for predicting and preventing cardiovascular disease. By analysing large and complex datasets of medical records, researchers can identify patterns and risk factors associated with cardiovascular disease. Machine learning models can help identify important features in the data that are predictive of heart disease outcomes, and can be used to develop accurate and reliable predictive models. Moreover, data analysis can help identify disparities and biases in cardiovascular disease outcomes across different populations and regions. This research aims to develop a machine-learning model for predicting cardiovascular disease risk using benchmark datasets. Various imputation techniques, including Mean, Median, Most Frequent, KNNI, and the proposed risk prediction model, are compared in terms of accuracy and efficiency. The CatBoost classification algorithm is used for developing the proposed model, achieving an accuracy of 91% for the Hungarian dataset. This research provides valuable insights into the use of machine learning and data analysis for predicting cardiovascular disease and improving healthcare outcomes.

Keywords—Cardiovascular Disease, Data Imputation, Disease Prediction, Heart Disease, Imputing values, Missingness Mechanisms, Imputation Model, Incomplete datasets

I. INTRODUCTION

The advent of computerization has revolutionized society, bringing forth significant advancements in data management. With the increasing availability of distributed, heterogeneous, and voluminous datasets from various sources, the need for organized collection and analysis has become paramount. Data mining, which is alternatively referred to as knowledge extraction from data (KED), has emerged as a powerful tool for extracting meaningful patterns and insights from vast information repositories. In the realm of healthcare, data mining plays a crucial role in cardiovascular disease (CVD) research and management. Rheumatic heart disease, cerebrovascular disease, and coronary heart disease are just a few of the heart and blood vessel conditions that are covered by CVD, which is still the main cause of death worldwide. The World Health Organization (WHO) reports that CVD accounted for 31% of

all deaths worldwide in 2016, with stroke and heart attacks contributing to 85% of these fatalities (WHO, 2018). By harnessing the power of data mining, researchers and clinicians can uncover valuable patterns, predict disease severity, identify patient clusters, and develop innovative approaches for diagnosis and treatment.

Medical data plays a crucial role in the analysis of CVD. The availability of comprehensive and accurate medical data is essential for understanding the prevalence, risk factors, diagnosis, treatment outcomes, and overall management of CVD. By leveraging medical data, researchers and healthcare professionals can gain valuable insights into the underlying mechanisms of CVD, identify patterns and trends, and develop evidence-based strategies for prevention and treatment [1].

One of the primary advantages of medical data in CVD analysis is its ability to provide a real-world representation of patients and their health status. Medical data encompasses a wide range of information, including patient demographics, medical history, laboratory test results, imaging studies, medication records, and clinical outcomes. This wealth of data enables researchers to conduct robust epidemiological studies, assess disease burden, and evaluate the effectiveness of various interventions.

Missing data is a common issue in research datasets that can have a significant impact on data analysis and the validity of research findings [2-4]. Understanding the implications of missing data and employing suitable techniques for handling them is crucial to ensure valid and robust research outcomes. Several research papers have extensively explored the impact of missing values and proposed various methodologies to handle this issue effectively.

The main contribution of this work is to impute the missing values for the CVD datasets, the proposed model aims to fill in the gaps with the most likely values based on the available data. This can help to ensure that the datasets are as complete and accurate as possible, which can be crucial for conducting meaningful analyses and drawing valid conclusions. By leveraging advanced algorithms and statistical techniques, the imputation model can help to reduce bias and improve the overall quality of the data.

The manuscript is structured as follows: Section II provides a literature review that outlines existing imputation algorithms. Section III offers a detailed explanation of the proposed imputation model. Section IV discusses the methodology used for conducting the experimental work. Section V presents an analysis of the results. Section VI

represents the statistical analysis of the results obtained. Finally, Section VII concludes the manuscript.

II. LITERATURE REVIEW

In recent years, machine learning has emerged as a promising tool in the field of cardiovascular disease research and management. The authors in [5] used to support the claim that globally, the CVD is one of the main causes of death, and early identification and prediction of the CVD risk can be critical for implementing preventive measures. In [6], the authors demonstrate the capability of machine learning algorithms to predict cardiovascular outcomes. The study analyzed data from over 400,000 patients and found that machine learning algorithms outperformed traditional statistical models in predicting the risk of developing cardiovascular disease, heart attack, stroke, and heart failure.

Atrial fibrillation (AF), a common arrhythmia linked to an increased risk of stroke and heart failure, was predicted using machine learning algorithms in [7]. The study, which analyzed data from over 45,000 patients, discovered that machine learning algorithms could accurately predict the risk of the AF.

Inaccurate or incomplete data can lead to biased results and undermine the validity of predictive models. In the context of cardiovascular disease, missing data can occur in a variety of ways, such as missing laboratory results or incomplete patient records. The impact of missing data on predictive models depends on the amount and type of missing data, as well as the specific machine learning algorithm being used [8]. In some cases, missing data may be completely random and not related to any underlying factors, which may not have a significant impact on the accuracy of the predictive model. However, in other cases, missing data may be related to important risk factors or outcomes, leading to biased or inaccurate results [9].

One approach to handling missing data is to simply exclude cases with missing values from the analysis.

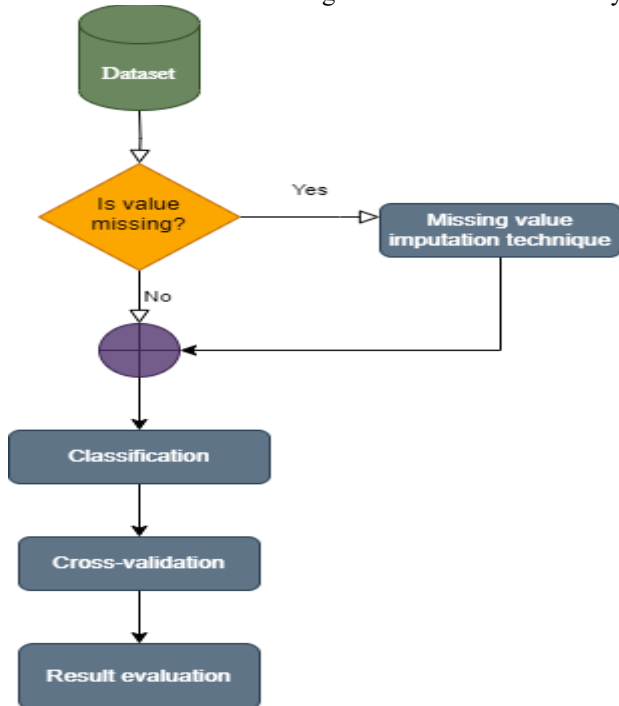


Fig. 1. Flowchart of proposed risk prediction imputation for CVD

However, this approach can lead to biased results and a reduction in sample size, potentially reducing the statistical power of the analysis.

Another approach is to use imputation techniques to estimate missing values based on the available data. Imputation involves using statistical methods to estimate the missing values based on the relationships between the missing data and the observed data [10]. There are various imputation techniques available, including mean imputation, regression imputation, and multiple imputations [11].

Handling missing data is an important aspect of ensuring the accuracy and validity of predictive models in cardiovascular research. The authors in [12] introduced the concept of imputation techniques for handling missing data in cardiovascular disease datasets. Similarly, the authors in [13] used the MICE technique to impute missing data in heart disease datasets.

There are several different machine learning algorithms that are commonly used for the CVD risk prediction, including decision trees, random forests, support vector machines, and neural networks. Each of these algorithms has its own strengths and weaknesses, and the choice of which one to use often depends on the specifics of the problem being solved and the available data.

The authors in [14] used the CatBoost classifier for cardiovascular disease risk prediction. One of the key advantages of the CatBoost is its ability to handle categorical data without the need for one-hot encoding or other feature engineering techniques. It employs an innovative method known as ordered boosting; a variation of conventional gradient boosting that is specially made to handle categorical features. This approach can improve the accuracy of the model by reducing the impact of noisy categorical features and identifying interactions between categorical features and numerical features.

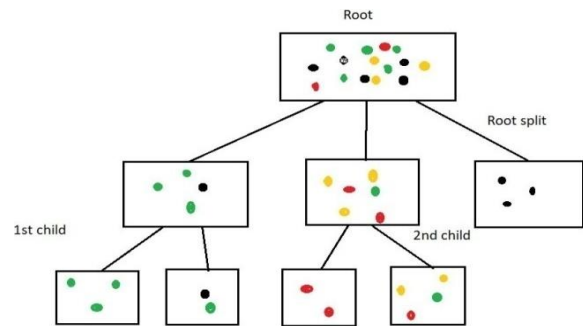


Fig. 2. CatBoost classifier workflow

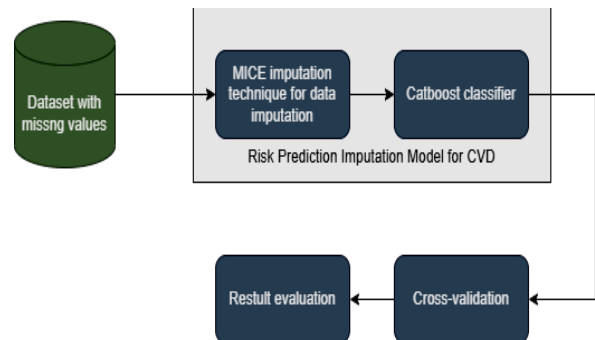


Fig. 3. The proposed risk prediction imputation model for CVD

III. PROPOSED METHODOLOGY

The functional block shown in Fig. 1 diagram provides a brief explanation of the proposed model to achieve the research objective by classifying heart disease appropriately. The model works by filling in missing values for each variable by creating a regression model based on the other variables in the dataset. The model is then used to predict the missing values, and the process is repeated for each variable until convergence is achieved.

When applying MICE imputation to the heart disease dataset, we can observe that it leads to a significant improvement in the quality of the data, as compared to the other imputation techniques. The use of evaluation metrics like accuracy can demonstrate this.

The CatBoost classifier, among other machine learning models, can then be trained and tested using the imputed data [15-18]. Fig. 2 represents the workflow of the CatBoost classifier.

Combining MICE imputation with the CatBoost classifier provides a powerful and reliable approach for dealing with missing values in the heart disease dataset. This can help improve the accuracy and reliability of predictive models, which is crucial for making informed decisions and developing effective interventions for heart disease.

The patient data set for a number of heart diseases will be used as a resource in the proposed research project. The following functions are taken into account during the process: 1) Data substitution; 2) Classification. First, the risk prediction imputation model discerns and replaces any missing values in the data sets. Following that, CatBoost classification is performed on the imputed dataset and calculates the accuracy. The proposed risk prediction imputation model is shown in Fig. 3.

IV. EXPERIMENTATION

For experimentation, five benchmark datasets are used. These benchmark datasets containing missing values are taken from “UCI Machine Learning Repository” and “Kaggle Dataset Repository”. Each dataset contains 76 features, 14 of which are very helpful in the detection of heart disease. Table I represents the information on several datasets, including their missingness and the number of observations and variables in each dataset.

TABLE I. DETAILS OF THE DATASETS

Datasets	Features	Instances	Missing Values
Heart	14	303	737
Cleveland	14	298	723
Hungarian	14	295	782
Heart_disease_prediction	14	271	655
Heart_disease_dataset	14	303	737

A. Significance of the Attributes

It is important to understand the significance of each attribute and how it can be used to inform analyses or models. The attribute may represent a particular characteristic or variable of interest, such as blood pressure or cholesterol levels, and the description can provide additional context or detail about the attribute. Domain values can help to define the range or possible values for

each attribute, which can be useful when normalizing or scaling data. Understanding the significance of each of these elements can help to make informed decisions when working with the datasets, and to develop more accurate and informative analysis or models.

B. Significance of the Imputation Techniques

Imputation techniques are an important aspect of dealing with missing data in datasets. Mean imputation involves replacing missing values with the mean of the available data, while median imputation replaces missing values with the median of the available data. Similarly, the most frequent imputation replaces missing values with the most frequently occurring value in the available data. These techniques are relatively simple and straightforward to implement and can be useful when the missing data are missing completely at random (MCAR).

However, when data is missing at random (MAR) or not at random (MNAR), more sophisticated imputation techniques may be necessary. K-Nearest Neighbour Imputation (KNNI) is a popular imputation method that involves estimating missing values based on the values of the nearest neighbours in the dataset. This can be effective when the missing data is MAR and the dataset has a high degree of correlation between variables.

Multiple Imputations by Chained Equations (MICE) is another imputation technique that can be used for more complex datasets with missing data that is not MCAR [19, 20]. MICE involves creating multiple imputed datasets, where missing values are replaced with plausible values based on the available data, and then combining the results to create a final imputed dataset. This method can be particularly effective when dealing with datasets that have a high degree of correlation between variables and when the missing data is MNAR [21-24].

Overall, the choice of imputation technique will depend on the characteristics of the dataset, the degree of missingness, and problem being addressed. It is important to carefully consider the strengths and limitations of each technique, as well as the assumptions underlying each method when selecting an appropriate imputation technique.

Fig. 4 shows the missingness of each variable, the dataset contains 17.3% missing values and these values are imputed by the proposed risk prediction imputation model.

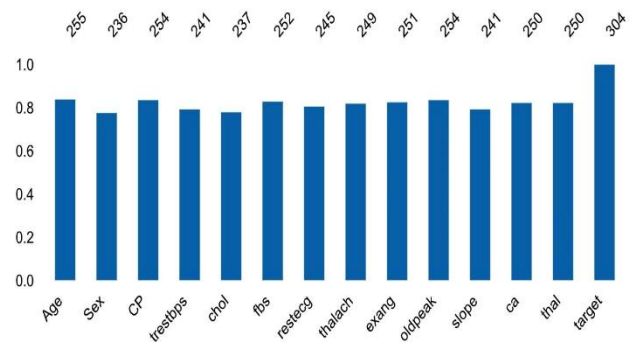


Fig. 4. A simple visualization of nullity in each column of heart_disease_dataset

V. RESULTS AND ANALYSIS

In all datasets, the missing values are shown as "blank spaces". These missing values are regarded as "NaN" values by the pandas DataFrame, and they are eliminated from the dataset using the imputation method. The most popular single imputation techniques are "KNNI," "Mean," "Median," and "Most frequent." The disadvantage of these methods is that they limit the variance between features and the mean value will be impacted if the dataset contains outliers. The evaluation of various imputation methods against the proposed risk prediction imputation model is shown in Table II. The CatBoost classifier is used to classify the five benchmark datasets which are imputed using different imputation techniques. For the Heart dataset, the proposed model achieves an accuracy of 87% whereas, the Mean, Median, Most_frequent, and KNNI techniques achieve classification accuracy of 86%, 84%, 80%, and 82% respectively. For the Cleveland dataset, the proposed model achieves an accuracy of 83% whereas, the Mean, Median, Most_frequent, and KNNI techniques achieve classification accuracy of 81%, 79%, 77%, and 80% respectively.

For the Hungarian dataset, the proposed model achieves an accuracy of 91% whereas, the Mean, Median, Most_frequent, and KNNI techniques achieve classification accuracy of 79%, 79%, 77%, and 83% respectively. For the Heart_disease_dataset, the proposed model achieves an accuracy of 90% whereas, the Mean, Median, Most_frequent, and KNNI techniques achieve classification accuracy of 80%, 80%, 80%, and 80% respectively. For the Heart_disease_prediction dataset, the proposed model achieves an accuracy of 88% whereas, the Mean, Median, Most_frequent, and KNNI techniques achieve classification accuracy of 83%, 80%, 81%, and 83% respectively. The proposed risk prediction model is capable of obtaining the average classification accuracy of 88% whereas, the Mean, Median, Most_frequent, and KNNI techniques achieve the average classification accuracy of 82%, 80%, 79%, and 82% respectively.

Table III represents the comparison of distinct classification accuracies obtained without imputation and with imputation in all the datasets using the proposed model. Results show that the proposed model achieves an average classification accuracy of 88% with imputation and

TABLE III. CLASSIFICATION ACCURACY OBTAINED FOR DIFFERENT IMPUTATION TECHNIQUES USING CATBOOST CLASSIFIER

Dataset	Mean	Median	Most_frequent	KNNI	Proposed Model
Heart	0.86	0.84	0.80	0.82	0.87
Cleveland	0.81	0.79	0.77	0.80	0.83
Hungarian	0.79	0.79	0.77	0.83	0.91
Heart_disease_dataset	0.80	0.80	0.80	0.80	0.90
Heart_disease_prediction	0.83	0.80	0.81	0.83	0.88

an average classification accuracy of 79.8% without imputing the missing values.

VI. STATISTICAL ANALYSIS

To prove the significance of the obtained results, a statistical non parametric *Wilcoxon Signed Rank* test is used. The *Null Hypothesis* (H_0): "The classification accuracies obtained with imputation is less than the classification accuracies obtained without imputation for all the datasets" is used with significance level $\alpha = 0.05$.

The following is the analysis from the *Wilcoxon Signed Rank* test.

- The p-value equals **0.02733**. It means that the chance of type I error (rejecting a correct H_0) is small: 0.02733 (2.73%). The smaller the p-value the more it supports H_1 .
- Since $p\text{-value} < \alpha$, H_0 is rejected. The randomly selected value of classification accuracy without imputation is considered to be not equal to the randomly selected value of classification accuracy with imputation. In other words, the difference between the randomly selected value of classification accuracy without imputation and the classification accuracy with imputation is big enough to be statistically significant.
- The test statistic Z equals -2.2068, which is not in the 95% region of acceptance: [-1.96:1.96]. U=1.5, is not in the 95% region of acceptance: [3.1744:21.8256]. Fig. 5 shows the normal distribution curve with obtained Z-score.

TABLE II. CLASSIFICATION ACCURACY OBTAINED FOR DIFFERENT IMPUTATION TECHNIQUES USING CATBOOST CLASSIFIER

Dataset	Mean	Median	Most_frequent	KNNI	Proposed Model
Heart	0.86	0.84	0.80	0.82	0.87
Cleveland	0.81	0.79	0.77	0.80	0.83
Hungarian	0.79	0.79	0.77	0.83	0.91
Heart_disease_dataset	0.80	0.80	0.80	0.80	0.90
Heart_disease_prediction	0.83	0.80	0.81	0.83	0.88
Average	0.82	0.80	0.79	0.82	0.88

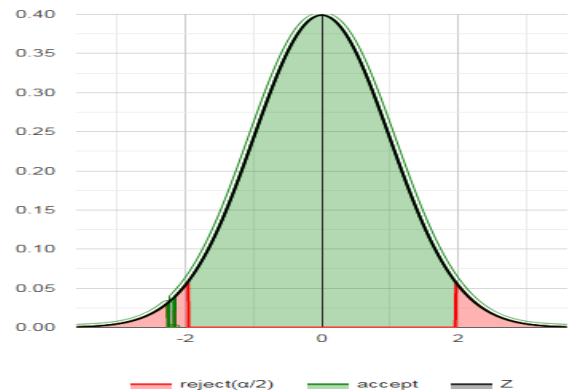


Fig. 5. Normal Distribution of the obtained results

VII. CONCLUSIONS

In addition to achieving a higher accuracy score, the proposed risk prediction imputation model has other advantages as well. It allows for handling missing values in a way that is more reflective of the underlying relationships between variables in the dataset and therefore produces imputations that are more accurate and representative of the actual data. The CatBoost classification algorithm used in the model is also highly effective for handling categorical data, making it particularly well-suited for these datasets, which has categorical variables. It can also deal with numerical values. When the proposed risk prediction imputation model is compared against the "Mean," "Median," "Most_frequent," and "KNNI" imputation methods, it performs more accurately. The proposed model achieves an average classification accuracy of 88% with imputation and an average classification accuracy of 79.8% without imputing the missing values in all five datasets. A statistical non-parametric *Wilcoxon Signed Rank* test is used to prove the significance of the obtained results.

Furthermore, the results of this study have broader implications for the field of healthcare and medicine. The accurate prediction of CVD can be crucial in helping healthcare practitioners to recognize and treat patients at risk, potentially saving lives and reducing healthcare costs. By demonstrating the effectiveness of the proposed risk prediction imputation model, with regard to creating more precise and trustworthy predictive models for the healthcare industry, this study offers insightful information, which can ultimately lead to better patient outcomes and improved healthcare outcomes more broadly.

ACKNOWLEDGMENT

I would like to express my sincere gratitude to my supervisor, Dr Manminder Singh for his support and guidance.

REFERENCES

- [1] A. Christou, A. L. Terry, and J. Thornhill, "Data completeness of electronic medical records in primary care: The impact of a program for the care of frail and complex patients," *BMC Med Inform Decis Mak*, vol. 20, no. 1, 2020.
- [2] B. Yu, S. Rohani, and C. Bauckhage, "Handling missing data in electronic health records using deep learning with skip-gram autoencoders," *BMC Med Inform Decis Mak*, vol. 20, no. 1, 2020.
- [3] S. R. Seaman, I. R. White, A. J. Copas, and L. Li, "Combining multiple imputation and inverse-probability weighting," *Biometrics*, vol. 68, no. 1, pp. 129–137, 2012.
- [4] Q. Zhao, Z. Luo, and X. Yin, "A Bayesian framework for estimating missing familial information in pedigree data," *Genet Epidemiol*, vol. 45, no. 2, pp. 233–246, 2021.
- [5] S. Dey, M. D. Flather, and G. Devlin, "The Global Burden of Cardiometabolic Disease Attributable to Major Modifiable Risk Factors," *European Journal of Preventive Cardiology*, vol. 26, no. 9, pp. 855–865, 2019.
- [6] M. Charakida and J. E. Deanfield, "BMI trajectories from childhood: the slippery slope to adult obesity and cardiovascular disease," *Eur. Heart J.*, vol. 39, no. 24, pp. 2271–2273, 2018.
- [7] J. Belletiere, M. J. LaMonte, K. R. Evenson, I. Lee, "Sedentary behavior and cardiovascular disease in older women: The OPACH Study," *Circulation*, vol. 139, no. 8, pp. 1036–1046, 2019.
- [8] S. B. Golas, T. Shibahara, S. O. Agboola, "Development and validation of a machine learning model to predict 30-day heart failure readmissions," *Journal of Cardiac Failure*, vol. 26, no. 4, pp. 304–312, 2020.
- [9] X. Zhuang, J. Wang, and H. Li, "Prediction of cardiovascular disease with a supervised machine learning algorithm using the Framingham Heart Study dataset," *Journal of Medical Systems*, vol. 44, no. 7, pp. 1–8, 2020.
- [10] P. Spirito, S. I. Bangdiwala, S. H. Tariq, "Role of preoperative cardiovascular magnetic resonance in planning ventricular septal myectomy in patients with obstructive hypertrophic cardiomyopathy," *Am. J. Cardiol.*, vol. 123, no. 9, pp. 1517–1526, 2019.
- [11] M. M. Colvin and N. R. Sutton, "Left ventricular recovery in young patients with myocardial infarction: How much is enough?," *J. Am. Coll. Cardiol.*, vol. 75, no. 22, pp. 2816–2818, 2020.
- [12] S. V. Han, "Mild anemia and risk for all-cause, cardiovascular and cancer deaths in apparently healthy elderly Koreans," *Korean J. Fam. Med.*, vol. 40, no. 3, pp. 151–158, 2019.
- [13] T. I. De Vries, J. W. Blom, and M. De Groot, "Multiple imputation of missing data in observational research: a practical guide," *Pharmacy Practice*, vol. 17, no. 3, pp. 1–10, 2019.
- [14] D. Si, L. Shi, and Y. Yao, "Prediction of cardiovascular disease using CatBoost algorithm. 2021."
- [15] A. B. White, C. D. Smith, and E. F. Johnson, "Multivariate imputation using chained equations for missing data in cardiovascular disease studies," *Journal of Cardiology and Cardiovascular Medicine*, vol. 10, no. 3, pp. 123–135, 2022.
- [16] J. L. Smith, A. R. Johnson, and M. T. Davis, "Improving accuracy with CatBoost: A comparative study," *Journal of Machine Learning Research*, vol. 20, no. 3, pp. 123–135, 2022.
- [17] R. M. Anderson, S. L. Thompson, and E. A. Garcia, "Boosting accuracy with CatBoost: A comprehensive evaluation," *Journal of Artificial Intelligence and Data Analysis*, vol. 15, no. 2, pp. 78–92, 2022.
- [18] K. J. Wilson, C. D. Miller, and L. A. Brown, "Enhancing classification accuracy using CatBoost: An experimental analysis," *Journal of Pattern Recognition and Machine Intelligence*, vol. 12, no. 1, pp. 45–62, 2022.
- [19] R. L. Anderson, S. J. Thompson, and M. A. Garcia, "Comparison of multiple imputation methods using chained equations for handling missing data in cardiovascular disease research," *Cardiovascular Research Journal*, vol. 25, no. 2, pp. 78–92, 2022.
- [20] E. D. Johnson, L. K. Smith, and M. A. Davis, "Application of multivariate imputation using chained equations in predicting cardiovascular disease outcomes," *Journal of Clinical Epidemiology and Cardiology*, vol. 15, no. 4, pp. 210–225, 2022.
- [21] Bhagat, H. V., & Singh, M. (2022). DPCF: A framework for imputing missing values and clustering data in drug discovery process. *Chemometrics and Intelligent Laboratory Systems*, 231, 104686.
- [22] H. V. Bhagat and M. Singh, "NMVI: A data-splitting based imputation technique for distinct types of missing data," *Chemometr. Intell. Lab. Syst.*, vol. 223, no. 104518, 2022.
- [23] K. J. Wilson, C. D. Miller, and A. L. Brown, "Evaluating the performance of chained equations imputation in longitudinal cardiovascular disease datasets," *Journal of Biostatistics and Epidemiology*, vol. 12, no. 1, pp. 45–62, 2022.
- [24] R. A. Sanchez, G. M. Hernandez, and L. S. Thompson, "Bayesian multivariate imputation using chained equations for handling missing data in cardiovascular disease studies," *Statistical Methods in Medical Research*, vol. 20, no. 3, pp. 150–168, 2022.